power

File: power.kicad_sch

logic

File: logic.kicad_sch

tubes

File: tubes.kicad_sch

Sheets: power, logic, tubes. Inter-sheet nets are global labels. Values marked _open_ are listed in docs/hardware.md (Nixie board electrical design).

Generated by tools/gen_nixie_schematics.py; values per docs/hardware.md

universal-nixie-display (open hardware)

Sheet: /

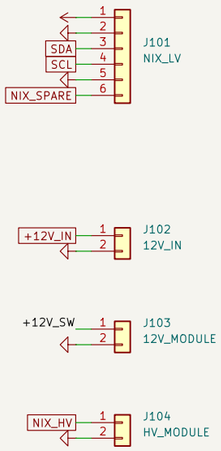
File: nixie.kicad_sch

Title: Nixie controller / mount

Size: A4	Date:	**Rev: Rev A draft**
KiCad E.D.A. 10.0.5		Id: 1/4

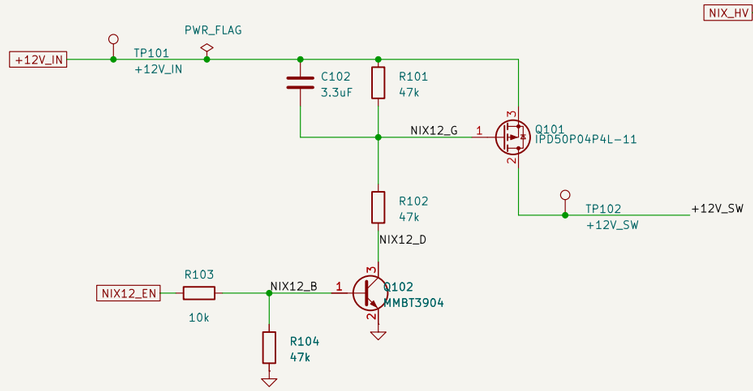
HARNESS LANDINGS (D14, soldered)

NIX_LV and NIX_12V come from the dekatron board; J103 and J104 go to the Nixie HV module (its 12 V input and its HV output).



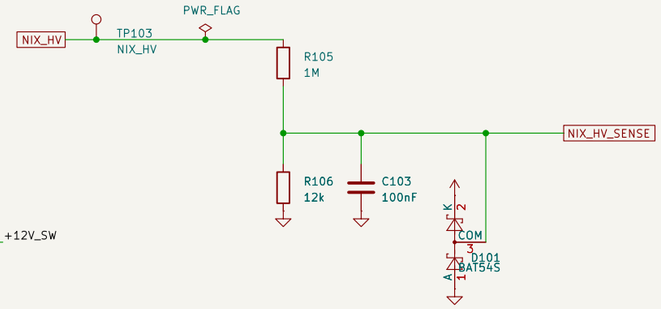
NIXIE HV MODULE 12 V SWITCH

High side, off by default: R101 holds Q101 off, R104 holds Q102 off while the RP2040 is in reset. C102 with R101 / R102 sets a soft start into the module's 470 uF (hardware/nixie/simulation/nix12_switch.cir).



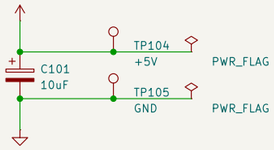
NIXIE HV MONITOR

Divider to ADC0: 3.3 V at about 278 V. R105 is also the bleeder (below 30 V in about 2–3 s, measured). BAT54S clamp and reservoir as on the dekatron ADC inputs.



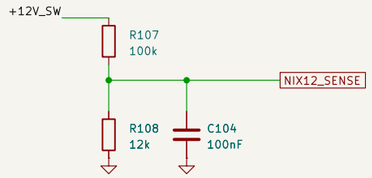
+5V BULK, TEST PADS

At the NIX_LV landing



SWITCHED 12 V SENSE

Divider to ADC1 (GP27), as the dekatron VIN_SENSE: 1.29 V at 12 V, 2.57 V at the 24 V clamp. Tells a switch fault from a module fault when NIX_HV is missing (user, 2026-11-11).



Generated by tools/gen_nixie_schematics.py; values per docs/hardware.md

universal-nixie-display (open hardware)

Sheet: /power/

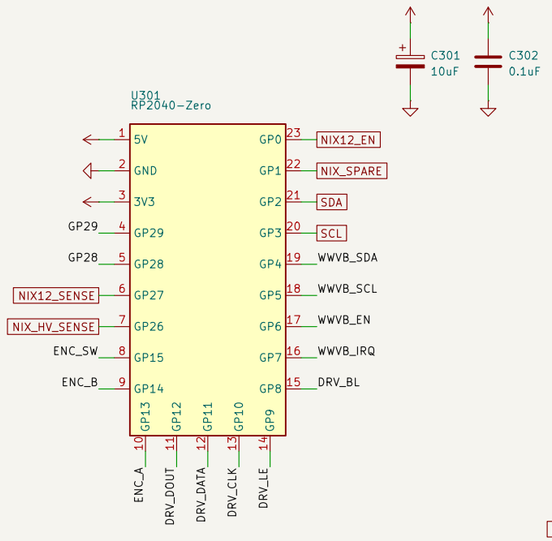
File: power.kicad_sch

Title: Power: harness landings, Nixie HV module 12 V switch, HV monitor

Size: A3	Date:	**Rev: Rev A draft**
KiCad E.D.A. 10.0.5		Id: 2/4

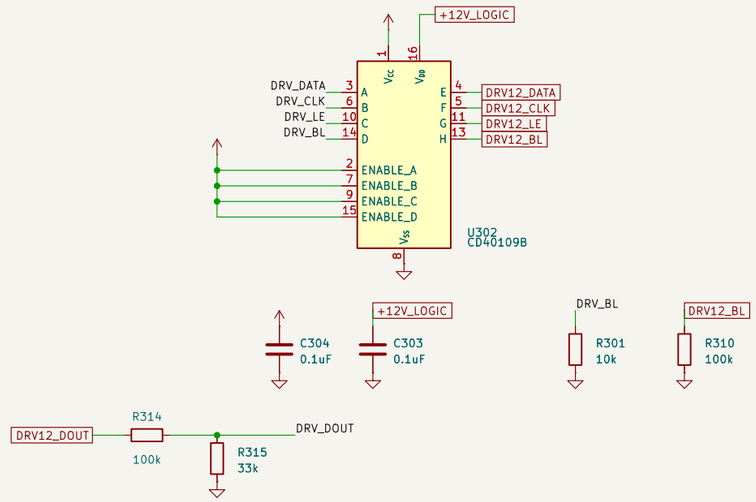
RP2040-Zero (pin map: docs/hardware.md)

Castellated, soldered by its edge pads; powered from +5V (NIX_LV). C301 / C302 at the module.



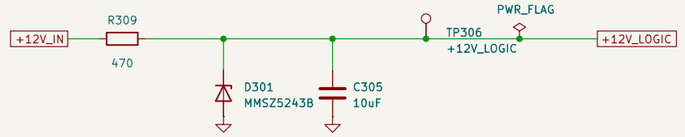
LEVEL SHIFTER: CD40109B, 3.3 V -> 12 V

HV5622 inputs need V_IH >= VDD - 2 V. Enables tied to 3V3. DRV_BL pulled down: all tube outputs off while the RP2040 is in reset (BL low, POL high). R310: the same while 3V3 is absent. Readback: U203 DOUT (DRV12_DOUT) to GP12 (SPI1 RX) through R314 / R315 (user, 2026-11-11).



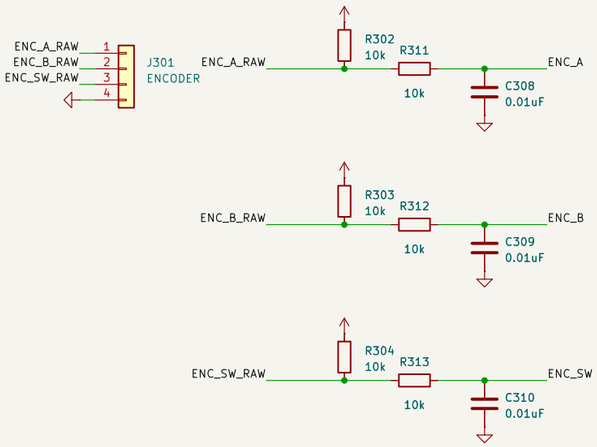
12 V LOGIC RAIL

HV5622 / CD40109B VDD. The HV5622's VDD absolute maximum is 15 V, below the 24 V TVS clamp on the 12 V input: series resistor and 13 V zener (12.35-13.65 V).



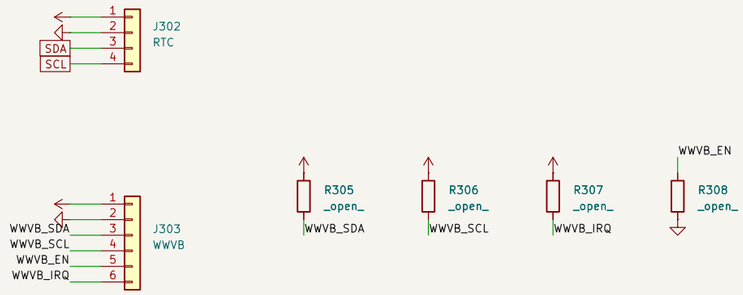
ENCODER (front panel, harness)

Bourns PEC11R wired to J301 (user, 2026-11-07). Filter per its data sheet (Suggested Filter Circuit): 10k pull-up, 10k series, 0.01 uF; the push switch the same.



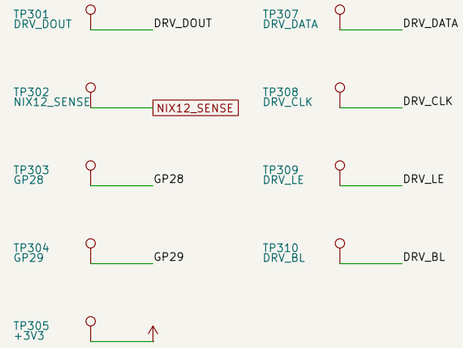
TIME MODULES (optional, harnesses)

RTC (ZS-042) on the main bus; ES100 on its private bus. Pull resistors here so an unplugged module leaves defined levels; values chosen at assembly.



TEST PADS

Readback, 12 V sense, spare GPIOs, 3V3, driver lines (3.3 V side)



Generated by tools/gen_nixie_schematics.py; values per docs/hardware.md

universal-nixie-display (open hardware)

Sheet: /logic/
File: logic.kicad_sch

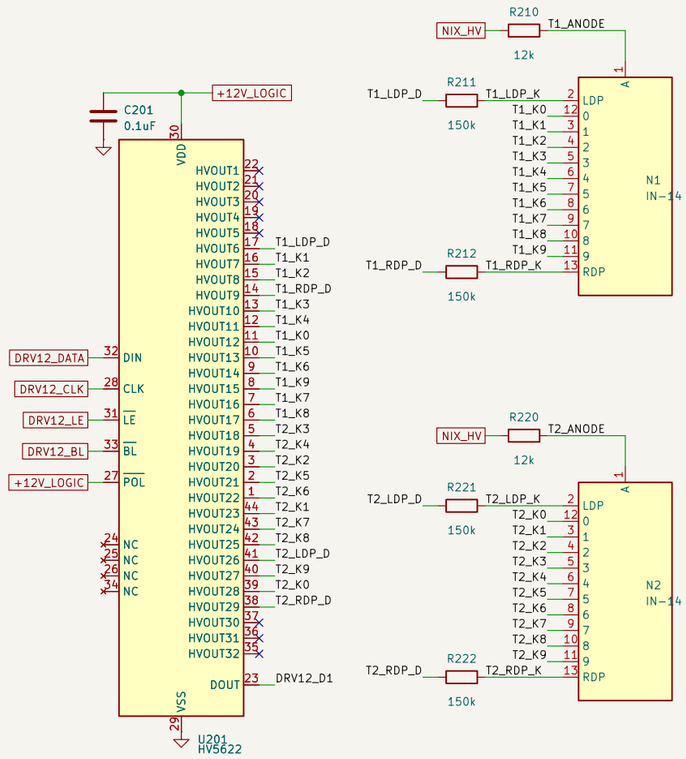
Title: Logic: RP2040-Zero, level shifter, encoder, time modules

Size: A3	Date:	**Rev: Rev A draft**
KiCad E.D.A. 10.0.5		Id: 3/4

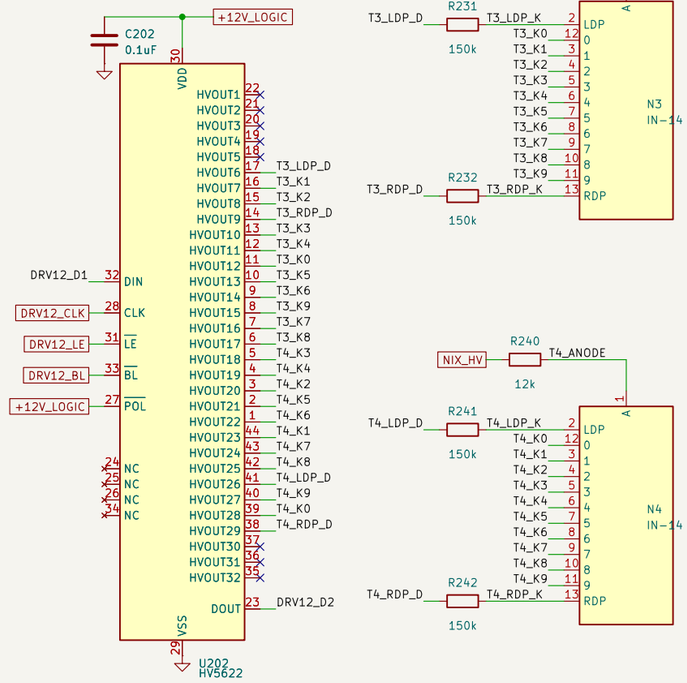
TUBE DRIVERS: 3 x HV5622 (220 V open drain)

Tube 1 at the left as seen by the viewer. Chain U201 -> U202 -> U203 (DRV12_D1, DRV12_D2), 96 bits latched whole (LE); U203 DOUT back to the RP2040 (DRV12_DOUT, readback). Outputs assigned from the PCB geometry (HV_MAP in tools/gen_nixie_schematics.py, table in docs/hardware.md): HVOUT1–5 and 30–32 spare. Firmware owns the map. Each tube: 12k anode resistor from NIX_HV, 150k in series with each decimal point.

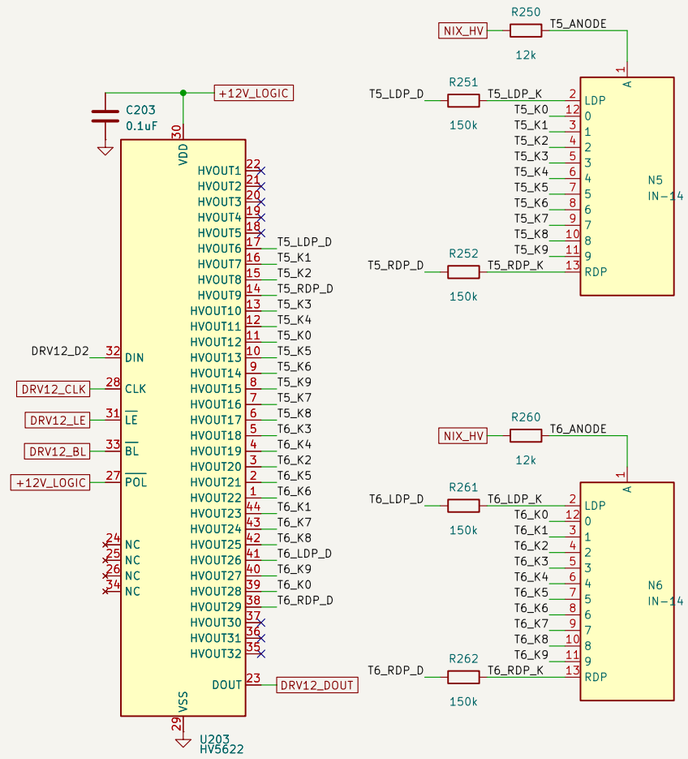
U201: TUBES 1 AND 2



U202: TUBES 3 AND 4



U203: TUBES 5 AND 6



Generated by tools/gen_nixie_schematics.py; values per docs/hardware.md

universal-nixie-display (open hardware)

Sheet: /tubes/
File: tubes.kicad_sch

Title: Tubes: HV5622 drivers, IN-14s, resistors

Size: A3	Date:	**Rev: Rev A draft**
KiCad E.D.A. 10.0.5		Id: 4/4